

Technical Audit Report

Formal Analysis of Mathematical Inconsistencies in the Fast Multipole Method Documentation

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Fast Multipole Methods for N-Body Problems*

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Prepared by: Monica AI Technical Review

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Abstract

This report presents the results of a formal technical audit of the documentation titled *Advanced Methods for Scientific Computing: Fast Multipole Methods for N-Body Problems*. A total of **14 distinct inconsistencies** were identified, spanning sign errors, normalization ambiguities, incomplete derivations, structural defects, and bibliographic insufficiencies. Each finding is documented with its precise location, a description of the problem, the correct mathematical form, and an assessment of its impact on implementation correctness. Findings are classified by priority (High, Medium, Low) and organised by mathematical section. A consolidated correction checklist is provided at the end of the report.

This document is intended for the authors and maintainers of the audited documentation. All mathematical formulas are reproduced from the source document for reference purposes.

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1 Executive Summary

The audited document provides a conceptually sound introduction to the Fast Multipole Method (FMM) for N-body problems, covering both two-dimensional (2D) and three-dimensional (3D) formulations together with an application to particle dynamics. The high-level algorithmic structure – tree construction, upward pass via M2M translations, interaction phase via M2L translations, downward pass via L2L translations, and leaf-level evaluation – is correctly described.

However, a systematic review of the mathematical content has revealed **14 inconsistencies** of varying severity. These range from critical sign errors that would cause incorrect forces in any implementation, to notation ambiguities that reduce readability, to structural defects that render entire sections unusable.

The 2D formulation contains important sign inconsistencies in the near-field evaluation and gradient computation. The 3D formulation introduces the correct mathematical objects but leaves translation formulas underdefined, with potentially incorrect summation limits and undefined symbols. The particle-dynamics chapter is incomplete, with a truncated time-integration section and corrupted equations.

Table 1 provides a consolidated overview of all 14 findings, their classification, and their priority level.

Table 1: Summary of all 14 identified inconsistencies.

#	Title	Section	Category	Priority
I-01	Sign error in 2D near-field potential	4.1	Sign error	High
I-02	Sign error in 2D far-field gradient	4.1	Sign error	High
I-03	Repeated sign error in electrostatic example	5.5.1	Sign error	High
I-04	Self-interaction exclusion not stated	General	Completeness	High
I-05	Incomplete time-integration section	5.3	Completeness	High
I-06	Corrupted equations in Chapter 5	5.5.1	Completeness	High
I-07	Undefined symbols in 3D translation formulas	4.2.1–4.2.4	Notation	High
I-08	Suspicious 3D L2L summation limits	4.2.4	Mathematical	High
I-09	3D spherical harmonic normalization unspecified	4.2.1	Mathematical	Medium
I-10	Condon-Shortley phase double-counting risk	4.2.1	Mathematical	Medium
I-11	Inconsistent M2L placement in algorithm	3.3, 3.5	Structural	Medium

Continued on next page

(Table 1 continued)

#	Title	Section	Category	Priority
I-12	Ambiguous translation vector directions in 3D	4.2.1–4.2.4	Notation	Medium
I-13	Inconsistent notation ϕ vs. Φ	Throughout	Notation	Low
I-14	Bibliographic insufficiency for 3D formulas	Bibliography	Bibliographi	Low

2 Scope and Methodology

The audit covers the full text of the document *Advanced Methods for Scientific Computing: Fast Multipole Methods for N-Body Problems*, including all mathematical derivations, algorithmic descriptions, and the bibliography. The review was conducted by systematic cross-checking of:

- (i) internal consistency of mathematical definitions and formulas,
- (ii) consistency between the 2D and 3D formulations,
- (iii) consistency between the algorithmic description and the mathematical derivations,
- (iv) completeness of all sections as listed in the table of contents,
- (v) correctness of summation limits and index conventions against standard FMM references, and
- (vi) adequacy of the bibliography for the technical claims made.

Each finding is assigned a unique identifier of the form **I-*nn***, a priority level, and a category. Priority levels are defined as follows.

- High** The error would cause incorrect numerical results in any implementation, or renders a section mathematically unusable. Must be corrected before the document can serve as an implementation reference.
- Medium** The error introduces ambiguity or a potential source of implementation mistakes, but the document remains partially usable. Should be corrected before release.
- Low** The issue reduces clarity or bibliographic rigour but does not directly affect mathematical correctness. Should be addressed in a revision.

3 High-Priority Findings

3.1 I-01 – Sign Error in the 2D Near-Field Potential

HIGH PRIORITY

Location: Section 4.1 (Two-Dimensional Formulation), near-field evaluation.
Category: Sign error.

Description

The 2D interaction kernel is defined in the document as

$$G(z, z_j) = -\log |z - z_j|, \quad (1)$$

and the corresponding physical potential is correctly stated as

$$\phi(z) = -\sum_j q_j \log |z - z_j|. \quad (2)$$

However, in the evaluation section, the near-field contribution is written as

$$\phi_i^{\text{near}} = \sum_{j \in N(i)} q_j \log |z_i - z_j|, \quad (3)$$

which is missing the leading negative sign.

Correct Form

The near-field contribution must be consistent with the kernel definition:

$$\phi_i^{\text{near}} = -\sum_{j \in N(i)} q_j \log |z_i - z_j|. \quad (4)$$

In real Cartesian coordinates this reads

$$\phi_i^{\text{near}} = -\sum_{j \in N(i)} q_j \log \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}. \quad (5)$$

Impact

The total potential is assembled as $\phi_i = \phi_i^{\text{near}} + \phi_i^{\text{far}}$. With the sign error in Eq. (3), the near-field and far-field contributions carry opposite sign conventions and cannot be consistently summed. Any implementation based on the erroneous formula will produce incorrect potentials and forces.

3.2 I-02 – Sign Error in the 2D Far-Field Gradient

HIGH PRIORITY

Location: Section 4.1, gradient/force evaluation subsection.

Category: Sign error.

Description

The auxiliary complex potential is defined as

$$U(z) = \sum_{j \in F(z)} q_j \log(z - z_j), \quad (6)$$

and the physical far-field potential is recovered as

$$\phi^{\text{far}}(z) = -\text{Re}[U(z)]. \quad (7)$$

The local expansion is evaluated as

$$W(z) = \sum_{n=0}^p L_n^{(T)}(z - z_T)^n, \quad \phi^{\text{far}}(z) = -\text{Re}[W(z)]. \quad (8)$$

The gradient of the far-field potential should therefore satisfy

$$\frac{\partial \phi^{\text{far}}}{\partial x} = -\text{Re}[W'(z)], \quad \frac{\partial \phi^{\text{far}}}{\partial y} = \text{Im}[W'(z)], \quad (9)$$

where

$$W'(z) = \sum_{n=1}^p n L_n^{(T)}(z - z_T)^{n-1}. \quad (10)$$

The document's gradient formula appears to omit the outer negative sign induced by $\phi^{\text{far}} = -\text{Re}[U]$.

Correct Form

The gradient vector and the corresponding force are

$$\nabla \phi^{\text{far}} = \begin{pmatrix} -\text{Re}[W'(z)] \\ \text{Im}[W'(z)] \end{pmatrix}, \quad F^{\text{far}} = -\nabla \phi^{\text{far}} = \begin{pmatrix} \text{Re}[W'(z)] \\ -\text{Im}[W'(z)] \end{pmatrix}. \quad (11)$$

Impact

A sign error in the force formula reverses the direction of the computed force on every particle, which is catastrophic for any particle-dynamics simulation.

3.3 I-03 – Repeated Sign Error in the Electrostatic Example

HIGH PRIORITY

Location: Section 5.5.1 (Electrostatic Interaction), 2D part.

Category: Sign error.

Description

The same sign error documented in I-01 recurs in the electrostatic example section. The near-field logarithmic contribution is again written with a positive sign:

$$\phi_i^{\text{near}} = \sum_{j \in N(i)} q_j \log |z_i - z_j|, \quad (12)$$

contradicting the kernel definition $G(z, z_j) = -\log |z - z_j|$ established in Section 4.1.

Correct Form

The expression must read

$$\phi_i^{\text{near}} = - \sum_{j \in N(i)} q_j \log |z_i - z_j|. \quad (13)$$

If the authors intend to use the opposite kernel convention $G(z, z_j) = +\log |z - z_j|$, then the entire 2D derivation – including the relationship between U and ϕ , the multipole expansion, and the force formula – must be revised consistently.

Impact

The recurrence of the same error in a different section suggests a systematic sign convention issue throughout the 2D formulation. Both occurrences must be corrected simultaneously.

3.4 I-04 – Self-Interaction Exclusion Not Stated

HIGH PRIORITY

Location: General N-body formulation, near-field evaluation (2D and 3D).

Category: Completeness.

Description

The document does not explicitly state that self-interactions must be excluded when the source and target particle sets are identical. The near-field sum is written as

$$\phi_i^{\text{near}} = \sum_{j \in N(i)} q_j G(x_i, x_j), \quad (14)$$

without the constraint $j \neq i$. For the 2D logarithmic kernel, $-\log |z_i - z_j|$ is singular; for the 3D Coulomb kernel, $1/|x_i - x_j|$ is also singular.

Correct Form

When sources and targets are the same set of particles, the near-field sum must be written as

$$\phi_i^{\text{near}} = \sum_{\substack{j \in N(i) \\ j \neq i}} q_j G(x_i, x_j). \quad (15)$$

Explicitly, for the 2D case:

$$\phi_i^{\text{near}} = - \sum_{\substack{j \in N(i) \\ j \neq i}} q_j \log |z_i - z_j|, \quad (16)$$

and for the 3D Coulomb case:

$$\phi_i^{\text{near}} = \sum_{\substack{j \in N(i) \\ j \neq i}} \frac{q_j}{|x_i - x_j|}. \quad (17)$$

Impact

Failure to exclude self-interactions causes singularities in the computed potential and force, leading to undefined or infinite values in any implementation.

3.5 I-05 – Incomplete Time-Integration Section

HIGH PRIORITY

Location: Section 5.3 (Time Integration).

Category: Completeness.

Description

Section 5.3 begins to introduce a time-stepping scheme with the notation

$$(x_i^n, v_i^n) \rightarrow \quad (18)$$

but the expression is left incomplete. The text then abruptly transitions to a discussion of interaction kernels and electrostatic examples, without specifying any complete integration algorithm.

Correct Form

The section should specify a concrete integration method. The minimum acceptable content is an explicit Euler scheme:

$$v_i^{n+1} = v_i^n + \Delta t \frac{F_i^n}{m_i}, \quad (19)$$

$$x_i^{n+1} = x_i^n + \Delta t v_i^n. \quad (20)$$

For a physically accurate simulation, the velocity Verlet (Störmer-Verlet) method is strongly recommended:

$$v_i^{n+1/2} = v_i^n + \frac{\Delta t}{2} \frac{F_i^n}{m_i}, \quad (21)$$

$$x_i^{n+1} = x_i^n + \Delta t v_i^{n+1/2}, \quad (22)$$

$$F_i^{n+1} = F_i(x_i^{n+1}), \quad (23)$$

$$v_i^{n+1} = v_i^{n+1/2} + \frac{\Delta t}{2} \frac{F_i^{n+1}}{m_i}. \quad (24)$$

The velocity Verlet scheme is symplectic, conserves energy to second order, and is the standard choice for N-body particle dynamics.

Impact

As written, Section 5.3 cannot be used to reproduce or implement a time-evolution algorithm. The particle-dynamics chapter is therefore incomplete as a technical reference.

3.6 I-06 – Corrupted Equations in Chapter 5

HIGH PRIORITY

Location: Section 5.5.1 (Electrostatic Interaction).

Category: Completeness.

Description

Multiple equations in the electrostatic interaction section are visibly incomplete or corrupted. For example, the 3D near-field potential appears as

$$\Phi^{\text{near}}(x_i) = \sum_{q \in N_i} \frac{q_q}{r_{iq}} = \sum_j q_q \cdots \quad (25)$$

where the expression stops mid-formula. Additionally, the derivation of the Cartesian gradient from spherical derivatives is missing intermediate steps, with the text jumping directly from angular derivatives to a rotation matrix without justification.

Correct Form

The 3D near-field potential should be written as

$$\Phi_i^{\text{near}} = \sum_{\substack{j \in N(i) \\ j \neq i}} \frac{q_j}{r_{ij}}, \quad r_{ij} = \|x_i - x_j\|. \quad (26)$$

The near-field gradient is

$$\nabla \Phi_i^{\text{near}} = - \sum_{\substack{j \in N(i) \\ j \neq i}} q_j \frac{x_i - x_j}{\|x_i - x_j\|^3}. \quad (27)$$

The force on a target charge Q_i is

$$F_i^{\text{near}} = -Q_i \nabla \Phi_i^{\text{near}} = Q_i \sum_{\substack{j \in N(i) \\ j \neq i}} q_j \frac{x_i - x_j}{\|x_i - x_j\|^3}, \quad (28)$$

assuming $q_j = Q_j/(4\pi\epsilon_0)$.

Impact

The corrupted equations prevent Section 5.5.1 from being mathematically reliable. The section must be reconstructed before it can serve as a reference.

3.7 I-07 – Undefined Symbols in 3D Translation Formulas

HIGH PRIORITY

Location: Equations around 4.90, 4.103, and 4.113 (Section 4.2.1–4.2.4).

Category: Notation / Completeness.

Description

The 3D multipole-to-multipole translation formula (approximately Eq. 4.90) reads

$$M_n^m = \sum_{j=0}^n \sum_{k=-j}^j O_{n-j}^{m-k} i^{|m|-|k|-|m-k|} A_j^k A_{n-j}^{m-k} \rho^j Y_j^{-k}(\alpha, \beta) / A_n^m, \quad (29)$$

where the following symbols are introduced without definition:

- O_{n-j}^{m-k} : the nature of this quantity (old multipole coefficient, translated coefficient, or another expansion type) is not stated;
- ρ, α, β : presumably the spherical coordinates of the translation vector, but this is never explicitly stated;
- A_n^m : partially defined elsewhere, but the complete definition and its relationship to the spherical harmonic normalization are not given.

Correct Form

Before each translation formula, the following must be defined:

1. the old expansion center c_{old} and the new expansion center c_{new} ;
2. the translation vector $d = c_{\text{new}} - c_{\text{old}}$;
3. the spherical coordinates (ρ, α, β) of d ;
4. the input coefficients (e.g., O_j^k as child multipole coefficients);
5. the output coefficients (e.g., M_n^m as parent multipole coefficients);
6. the normalization coefficient A_n^m with its complete formula.

Impact

Without these definitions, the 3D translation formulas cannot be verified or implemented. This is a critical deficiency for any reader attempting to use the document as an implementation reference.

3.8 I-08 – Suspicious 3D L2L Summation Limits

HIGH PRIORITY

Location: Section 4.2.4 (Local-to-Local Translation).

Category: Mathematical error.

Description

The 3D local-to-local (L2L) translation formula is given as

$$L_n^m(P \rightarrow C) = \sum_{k=0}^n \sum_{l=-k}^k S_{n-k}^{m-l}(d_{PC}) L_k^l(P). \quad (30)$$

This structure is inconsistent with the analogous 2D formula, which is stated correctly in the document as

$$L_k^{(C)} = \sum_{n=k}^p \binom{n}{k} L_n^{(P)} (z_C - z_P)^{n-k}. \quad (31)$$

In the 2D case, a child coefficient of degree k receives contributions from parent coefficients of degree $n \geq k$. The analogous 3D formula should have an upper-triangular structure: a child coefficient of degree n should receive contributions from parent coefficients of degree $k \geq n$, up to the truncation order p .

Correct Form

The expected structure of the 3D L2L translation is

$$L_n^m(C) = \sum_{k=n}^p \sum_l T_{n,m}^{k,l}(d_{PC}) L_k^l(P), \quad (32)$$

where the exact form of the translation kernel $T_{n,m}^{k,l}$ and the constraints on l depend on the chosen solid-harmonic convention. The key point is that the outer summation should run from $k = n$ to $k = p$, not from $k = 0$ to $k = n$.

Impact

If the summation limits are reversed, local expansions propagated down the tree will accumulate incorrect contributions, producing wrong potentials at all leaf cells.

4 Medium-Priority Findings

4.1 I-09 – 3D Spherical Harmonic Normalization Unspecified

MEDIUM PRIORITY

Location: Section 4.2.1 (Spherical Harmonics and Translation Operators).

Category: Mathematical ambiguity.

Description

The spherical harmonics are defined as

$$Y_n^m(\theta, \phi) = C_n^m P_n^{|m|}(\cos \theta) e^{im\phi}, \quad C_n^m = \sqrt{\frac{(n - |m|)!}{(n + |m|)!}}. \quad (33)$$

This normalization omits the factor $\sqrt{(2n+1)/4\pi}$ present in the fully orthonormal convention. While a non-standard normalization is not inherently incorrect, the M2M, M2L, and L2L translation coefficients depend critically on the chosen normalization. The document presents translation formulas without explicitly demonstrating that they are consistent with this particular normalization.

Correct Form

The document should either adopt the fully normalized convention

$$Y_n^m(\theta, \phi) = \sqrt{\frac{2n+1}{4\pi} \frac{(n-|m|)!}{(n+|m|)!}} P_n^m(\cos \theta) e^{im\phi}, \quad (34)$$

or retain the current convention but explicitly state it and verify that all translation formulas are derived under the same convention. The complex conjugation relation $Y_n^{-m} = (-1)^m \overline{Y_n^m}$ should also be stated explicitly.

Impact

An inconsistency between the normalization of Y_n^m and the translation coefficients will cause all 3D M2M, M2L, and L2L operations to produce incorrect results.

4.2 I-10 – Condon-Shortley Phase Double-Counting Risk

MEDIUM PRIORITY

Location: Section 4.2.1.

Category: Mathematical ambiguity.

Description

The associated Legendre polynomial is defined as

$$P_n^{|m|}(x) = (-1)^{|m|} (1-x^2)^{|m|/2} \frac{d^{|m|}}{dx^{|m|}} P_n(x), \quad (35)$$

which includes the Condon-Shortley phase $(-1)^{|m|}$. If later formulas for spherical harmonics or translation coefficients independently assume the presence or absence of this phase, sign errors will propagate through all expansion coefficients.

Correct Form

The document should explicitly state:

1. whether the Condon-Shortley phase is included in P_n^m ,
2. whether it is included again in Y_n^m , and
3. how this choice affects the translation coefficients.

A consistent convention must be maintained throughout all formulas.

Impact

Phase inconsistencies lead to incorrect signs and complex phases in the expansion coefficients, affecting all 3D translation operators.

4.3 I-11 – Inconsistent Placement of M2L in the Algorithm

MEDIUM PRIORITY

Location: Sections 3.3 (Interaction Phase) and 3.5 (Evaluation).

Category: Structural inconsistency.

Description

Section 3.3 correctly describes M2L as part of the interaction phase, performed for well-separated cells at each tree level. However, Section 3.5 states:

Once a leaf cell is reached, the M2L translation is performed with the leaf cell as the target cell and the source cells given by its interaction list.

This description implies that M2L translations are performed only at the leaf level, which is incorrect for a classical FMM. In a standard implementation, M2L interactions are computed level by level for all cells at the corresponding tree levels, and the resulting local expansions are then propagated downward by L2L translations.

Correct Form

A correct algorithmic description of the FMM interaction phase is:

1. Construct multipole expansions at the leaf cells.
2. Perform the upward pass: translate multipole expansions from children to parents using M2M.
3. For each level, compute M2L interactions between well-separated cells in the interaction list.
4. Accumulate local expansions at target cells.
5. Perform the downward pass: propagate local expansions from parents to children using L2L.
6. At the leaves, evaluate the accumulated local expansion at target particles.
7. Compute near-field interactions directly.

Impact

The current wording may lead to an implementation where M2L interactions are only performed at the leaf level, potentially missing valid far-field interactions at coarser levels, especially in adaptive trees.

4.4 I-12 – Ambiguous Translation Vector Directions in 3D

MEDIUM PRIORITY

Location: Sections 4.2.1–4.2.4.

Category: Notation ambiguity.

Description

The document defines translation vectors such as $d_{CP} = c_C - c_P$ and states that each displacement vector describes the relative position of the new expansion center with respect to the original one. However, for M2M translation from child C to parent P , the original center is c_C and the new center is c_P , so the new-relative-to-original vector would be $c_P - c_C$, which is the opposite of $d_{CP} = c_C - c_P$.

Correct Form

All translation vectors should be defined consistently. The recommended convention is

$$d_{\text{old} \rightarrow \text{new}} = c_{\text{new}} - c_{\text{old}}, \quad (36)$$

giving:

$$d_{C \rightarrow P} = c_P - c_C \quad (\text{M2M: child to parent}), \quad (37)$$

$$d_{S \rightarrow T} = c_T - c_S \quad (\text{M2L: source to target}), \quad (38)$$

$$d_{P \rightarrow C} = c_C - c_P \quad (\text{L2L: parent to child}). \quad (39)$$

Alternatively, the current sign convention may be retained, but every translation formula must then use the matching convention.

Impact

A sign error in translation vectors changes the phases of spherical harmonics and produces incorrect translated coefficients in all 3D operators.

5 Low-Priority Findings

5.1 I-13 – Inconsistent Notation for the Physical Potential

LOW PRIORITY

Location: Throughout the document.

Category: Notation.

Description

The document uses both ϕ and Φ for the potential without a clear and consistently applied distinction. The auxiliary complex potential U is also introduced in the 2D section. The resulting notation is ambiguous: it is not always clear whether a given formula refers to the mathematical normalized potential, the physical electrostatic potential, or the complex auxiliary potential.

Correct Form

A consistent notation should be adopted throughout. A recommended convention is:

$U(z)$: complex auxiliary potential (2D only), (40)

$\phi(x)$: normalized mathematical potential computed by the FMM, (41)

$\Phi(x)$: physical potential (e.g., electrostatic potential including constants). (42)

The relationship between them should be stated explicitly. For the 3D Coulomb case with $q_j = Q_j/(4\pi\epsilon_0)$:

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \sum_j \frac{Q_j}{|x - x_j|} = \sum_j \frac{q_j}{|x - x_j|} = \phi(x). \quad (43)$$

Impact

The notation inconsistency reduces readability and may cause confusion between the mathematical and physical interpretations of the computed quantity.

5.2 I-14 – Bibliographic Insufficiency for 3D Formulas

LOW PRIORITY

Location: Bibliography.

Category: Bibliographic.

Description

The document contains advanced 3D spherical harmonic translation formulas but does not provide sufficiently precise references for the exact formulas and normalization conventions used. The bibliography includes some useful references (notably Cheng, Greengard, and Rokhlin, 1999), but the 3D translation coefficient derivations are not explicitly linked to a verifiable source. Additionally, the radiative transfer reference (Korkin et al.) appears weakly connected to the FMM N-body context and is not cited in the main text.

Correct Form

The following references should be added and explicitly cited at the relevant equations:

1. **Greengard and Rokhlin (1987)**: the original FMM paper, for the foundational algorithm.
2. **Gumerov and Duraiswami**: for spherical harmonic translation operators and implementation details.
3. **White and Head-Gordon**: for rotation-based 3D FMM translations.
4. **Dehnen**: for multipole methods in gravitational dynamics.
5. **Cheng, Greengard, and Rokhlin (1999)**: already cited; should be explicitly linked to the adaptive 3D formulation.

Impact

Without precise references, the 3D translation formulas cannot be independently verified, and the document cannot be trusted as a self-contained implementation reference.

6 Consolidated Correction Checklist

The following checklist summarises all required corrections, grouped by priority. Items marked **High** must be resolved before the document can be used as an implementation reference.

High-Priority Corrections

- [H-1] (I-01) Fix the sign of the 2D near-field potential in Section 4.1: use $\phi_i^{\text{near}} = -\sum_{j \in N(i)} q_j \log |z_i - z_j|$.
- [H-2] (I-02) Correct the 2D far-field gradient formula to include the outer negative sign from $\phi^{\text{far}} = -\text{Re}[U]$.
- [H-3] (I-03) Fix the sign of the 2D near-field contribution in the electrostatic example (Section 5.5.1).
- [H-4] (I-04) Add the constraint $j \neq i$ to all near-field sums where source and target particles coincide.
- [H-5] (I-05) Complete Section 5.3 with a concrete time-integration scheme (at minimum explicit Euler; velocity Verlet recommended).
- [H-6] (I-06) Reconstruct the corrupted equations in Section 5.5.1, including the 3D near-field potential, gradient, and force formulas.
- [H-7] (I-07) Define all symbols (O_n^m , ρ , α , β , A_n^m) before each 3D translation formula.
- [H-8] (I-08) Verify and correct the 3D L2L summation limits; the outer sum should run from $k = n$ to $k = p$, not from $k = 0$ to $k = n$.

Medium-Priority Corrections

- [M-1] (I-09) State the spherical harmonic normalization convention explicitly and verify that all translation formulas are consistent with it.
- [M-2] (I-10) Clarify whether the Condon-Shortley phase is included in P_n^m , in Y_n^m , or in both, and ensure consistency throughout.
- [M-3] (I-11) Revise the algorithmic description to correctly state that M2L translations are performed level by level, not only at the leaf level.
- [M-4] (I-12) Define all translation vectors consistently using a single sign convention throughout Sections 4.2.1–4.2.4.

Low-Priority Corrections

- [L-1] (I-13) Adopt a consistent notation for ϕ , Φ , and U throughout the document, with explicit definitions.
- [L-2] (I-14) Add targeted references for the 3D translation formulas and remove or contextualise the weakly connected radiative transfer reference.

7 Overall Assessment

Table 2: Distribution of findings by priority and category.

Category	High	Medium	Low	Total
Sign error	3	0	0	3
Completeness	3	0	0	3
Notation	1	1	1	3
Mathematical	1	2	0	3
Structural	0	1	0	1
Bibliographic	0	0	1	1
Total	8	4	2	14

The audited document provides a conceptually sound framework for the Fast Multipole Method and correctly conveys the high-level algorithmic structure. The 2D derivation is largely coherent, but it contains three sign errors (I-01, I-02, I-03) that are systematic and must be corrected together. The 3D formulation introduces the correct mathematical objects – spherical harmonics, regular and singular solid harmonics, and translation operators – but the translation formulas are insufficiently defined (I-07), potentially contain incorrect summation limits (I-08), and rely on an unspecified normalization (I-09, I-10). The particle-dynamics chapter is incomplete (I-05, I-06) and structurally deficient (I-11).

Of the 14 findings, 8 are classified as High priority, 4 as Medium priority, and 2 as Low priority (see Table 2). The document should not be used as an implementation reference until all High-priority findings have been resolved. After those corrections, the Medium-priority findings should be addressed to ensure the 3D section is mathematically rigorous. The Low-priority findings can be addressed in a final editorial pass.

References

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